

SOLE SOURCE JUSTIFICATION FOR ACQUISITION UNDER THE SIMPLIFIED ACQUISITION THRESHOLD

1. Contracting Agency and Activity. The Department of the Interior, United States Geological Survey, plans to contract on a sole source basis.

2. Nature of Action Being Approved.

New/Follow-On Purchase Order to Geotek Ltd. for the procurement of a micro-XRF sensor attachment that is compatible with the GMEG Science Center-owned Geotek Ltd. MSCL-XZ (Mark II).

3. Description of Supplies or Services.

The USGS Geology, Minerals, Energy, & Geophysics Science Center (GMEGSC) in Moffett Field, CA, requires a micro-XRF sensor to mount on its pre-existing Geotek Ltd. Multi-Sensor Core Logger instrument to conduct high-resolution, non-destructive elemental analyses of split sediment and rock cores, enabling continuous geochemical profiling at sub-millimeter scales. This sensor will complement the existing sensor suite which includes a hyperspectral paired VNIR/SWIR camera system, a visible light linescan camera, and a magnetic susceptibility point sensor. The primary use will be to support two USGS Land Change Science Program-funded projects: the *USGS Pacific Abrupt Climate Event (PACE) Project*, and the *USGS Reconstructing Western US Climate Variability and Ecosystem Response to Inform Climate Change Adaptation Strategies Project*. Other mission-critical research projects will also utilize this equipment to assess past changes in ocean productivity, precipitation, river discharge, nutrient cycling, lake ecosystem changes, field mapping assessments to develop Critical Mineral Resources, environmental health studies of mine tailing drainage, and wildfire impacts on elemental cycling in burned landscapes.

4. Estimated Dollar Value.

5. Statutory Authority. The proposed action may be awarded on a sole source basis under the authority of FAR 12.102(a) - For acquisitions valued at or below the simplified acquisition threshold (SAT), document the decision that only one source is available and the basis for the decision.

6. Rationale Supporting Use of Citation in No. 5.

The Geotek Ltd. micro-XRF sensor attachment integrates exclusively with the Geotek Ltd. MSCL-XZ (Mark II) and provides unique elemental mapping synchronized with existing sensors. Geotek Ltd. is the only manufacturer of sediment core logging systems in the world that pair a micro-XRF geochemistry sensor with hyperspectral VNIR/SWIR imaging cameras mounted on an XYZ-movement sediment core scanner. As a result, the capabilities provided by the resultant Geotek MSCL-XZ-II system are unique and unavailable from any other manufacturer. The configuration of the Geotek MSCL-XZ-II permits simultaneous non-destructive imaging of mineral and organic compound contents, and the micro-XRF sensor will allow the collection of elemental geochemical data that complement these data types. Imaging and characterization of these contents is a mission-critical aspect of the USGS projects listed in #3 that is only available from Geotek Ltd.

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[REDACTED]

7. Other Information. N/A

8. The Efforts to Identify Additional Sources Including the Market Research Conducted.

After reviewing the scientific literature associated with micro-XRF of sediment core samples, several key company websites were identified and investigated—these companies either (1) do not pair additional sensor technologies with hyperspectral cameras (such as already mounted on the USGS-owned MSCL-XZ-II) into a single instrument, or (2) do not offer sediment core system-mounted micro-XRF sensors. Additionally, Sources Sought Notice 140G0326G0100 was posted on SAM.gov for four days and received no responses. Based on the results of this market research, it is concluded that Geotek Ltd. is the only company in the world that combines multiple sensor systems into multi-function sediment core logging systems.

9. Future Plans to Permit Competition.

Any future similar requirements will be competed at the maximum extent practicable.

10. Recommendation and Certification from Program Office

Based on the above, I recommend this acquisition be conducted on a sole source basis and certify the above statements are true and accurate.

[REDACTED]

[REDACTED] Research Geologist

Date

11. Certifications from the Contracting Officer:

- a. The contracting officer must include a determination that the anticipated cost to the Government will be fair and reasonable with a short statement for the basis of that determination.
- b. This justification is accurate and complete to the best of my knowledge and belief.

[REDACTED]

Yangzhi Deng, Contracting Officer

Date

12. Approvals

[REDACTED]

[REDACTED] Bureau Competition Advocate, acting

Date